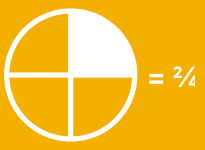
Equivalent fractions and simplifying



In this sheet you will…

- write equivalent fractions
- simplify a fraction using the HCF
- give two fractions the same denominator
- compare fractions

Key words

equivalent fractions
numerator · denominator
simplest form
HCF
common denominator

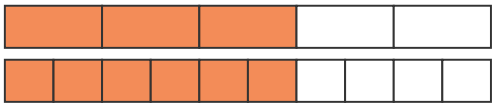
This sheet follows our lesson step by step, so you can read it again at home.

1 The same fraction with different names

$\frac{3}{5}$ and $\frac{6}{10}$ look different, but they describe the **same amount**:

3/5

6/10



Equivalent fractions
Fractions with the **same value**, written with different numbers.

How to make them
Multiply (or divide) the top **and** the bottom by the **same number**. You are multiplying by 1: $\frac{2}{2} = 1$, so the value does not change.

Worked example 1 · Fill in the missing numbers

$\frac{3}{4} = \frac{\square}{20}$ and $\frac{3}{4} = \frac{9}{\square}$

1a $\frac{3}{4} \overset{\times 5}{=} \frac{15}{20}$ — From 4 to 20 we multiply by 5, so we multiply the top by 5 too: 3 × 5 = 15.

1b $\frac{3}{4} \overset{\times 3}{=} \frac{9}{12}$ — From 3 to 9 we multiply by 3, so we multiply the bottom by 3 too: 4 × 3 = 12.

2 Simplifying a fraction

We can also **divide** top and bottom by the same number. A fraction is in **simplest form** when the top and bottom have no common factor except 1. To do it in one step, divide by the **HCF**.

Worked example 2 · Simplify $\frac{30}{48}$

Step 1 $30 = 2 \times 3 \times 5$
$48 = 2 \times 2 \times 2 \times 2 \times 3 = 2^4 \times 3$
Break both numbers down into prime factors.

Step 2 HCF = 2 × 3 = **6**
The shared primes are 2 and 3.

Step 3 $\frac{30}{48} \overset{\div 6}{=} \frac{5}{8}$
Divide top and bottom by 6. Check: 5 and 8 share nothing, so this is the simplest form.

Watch out!

Adding the same number to the top and the bottom does not give an equivalent fraction.
$\frac{2}{3} \to \frac{2+2}{3+2} = \frac{4}{5}$? No: $\frac{2}{3}$ is about 0.67 but $\frac{4}{5}$ is 0.8. We must **multiply or divide**, never add.

3 Comparing fractions

To compare two fractions, first give them the **same denominator**. If one denominator is a multiple of the other, **rewrite the other fraction**. Below are **two separate examples**: 3a and 3b.

Worked example 3a · Which is bigger, 3/5 or 7/10?

$\frac{3}{5}$ or $\frac{7}{10}$

Step 1 $10 = 5 \times 2$

10 is a multiple of 5, so we rewrite the **orange** fraction (the one with denominator 5).

Step 2 $\frac{3}{5} \overset{\times 2}{=} \frac{6}{10}$

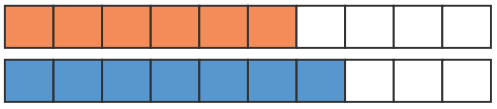
Multiply the top and the bottom by 2.

Step 3 $\frac{6}{10}$ and $\frac{7}{10}$: $6 < 7$

Same denominator now, so compare the numerators.

3/5 = 6/10

7/10



Answer: $\frac{3}{5} < \frac{7}{10}$. $\frac{7}{10}$ **is bigger.**

Worked example 3b · Which is bigger, 7/8 or 3/4?

$\frac{7}{8}$ or $\frac{3}{4}$

Step 1 $8 = 4 \times 2$

8 is a multiple of 4, so we rewrite the **orange** fraction (the one with denominator 4).

Step 2 $\frac{3}{4} \overset{\times 2}{=} \frac{6}{8}$

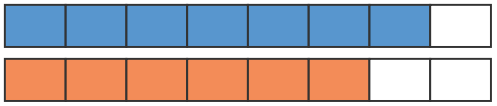
Multiply the top and the bottom by 2.

Step 3 $\frac{7}{8}$ and $\frac{6}{8}$: $7 > 6$

Same denominator now, so compare the numerators.

7/8

3/4 = 6/8



Answer: $\frac{7}{8} > \frac{3}{4}$. $\frac{7}{8}$ **is bigger.**

Key idea: comparing fractions

1. Make the denominators the **same**: rewrite the fraction with the smaller denominator.
2. Then compare the **numerators**: the bigger numerator is the bigger fraction.
3. **Tip:** ask "what do I multiply the small denominator by to reach the big one?" and use that number on the top too.

Exercise practice from the lesson

A **Fill in the missing number.**

1 $\frac{3}{8} = \frac{\square}{24}$

2 $\frac{5}{6} = \frac{20}{\square}$

3 $\frac{7}{9} = \frac{\square}{45}$

4 $\frac{18}{24} = \frac{\square}{4}$

B **Simplify fully.** Show the HCF.

1 $\frac{12}{18} =$ _____

2 $\frac{35}{42} =$ _____

3 $\frac{45}{60} =$ _____

4 $\frac{56}{72} =$ _____

C **Write the fractions with the same denominator, then compare them** (<, > or =).

1 $\frac{2}{3}$ and $\frac{5}{12}$

2 $\frac{3}{5}$ and $\frac{7}{20}$

3 $\frac{5}{6}$ and $\frac{11}{12}$

4 $\frac{3}{4}$ and $\frac{18}{24}$

Write your steps in your notebook.

D Priya says: "$\frac{3}{5} = \frac{5}{7}$ *because I added 2 to the top and the bottom.*" What is wrong? Find a fraction that is really equivalent to $\frac{3}{5}$.

? What do you think?

1 Is $\frac{3}{4}$ equal to $\frac{18}{24}$? How can you show it in **two** different ways?

2 What could we do to compare $\frac{2}{3}$ and $\frac{3}{4}$? Neither denominator is a multiple of the other. (We will see this in the next lesson.)

CHALLENGE

1. Find a fraction between $\frac{1}{3}$ and $\frac{1}{2}$. (Hint: use twelfths.) 2. Find three fractions equivalent to $\frac{4}{7}$ with a denominator under 40.

Summary checklist

- ☐ I can write equivalent fractions by multiplying or dividing top and bottom by the same number.
- ☐ I can simplify a fraction using the HCF.
- ☐ I can give two fractions the same denominator when one denominator is a multiple of the other, and compare them.